# Pythagorean Theorem: Mini-Lessons

NC.8.G.6–8

Name _____ Date _____ Period _____

## 5.1 Legs, hypotenuse and $a^2 + b^2 = c^2$ NC.8.G.6

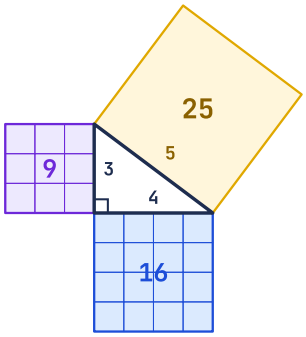
In a **right triangle**, the two shorter sides are the **legs** (a and b). The longest side, across from the right angle, is the **hypotenuse** (c). The theorem: **$a^2 + b^2 = c^2$**.

**Right triangle**
a triangle with a 90° angle (a square corner)

**Leg**
one of the two sides that make the square corner

**Hypotenuse**
the longest side, across from the square corner

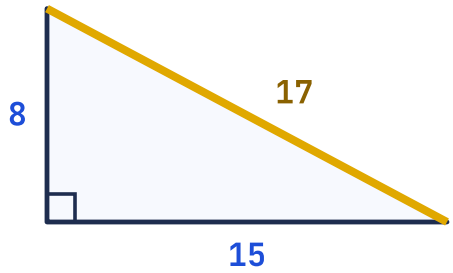
**Watch out:** The hypotenuse is **always** across from the square corner. Find it first.




$9 + 16 = 25$

$a^2 + b^2 = c^2$

### Walk through: Which side is the hypotenuse?




1. Where is the right angle (square corner)?

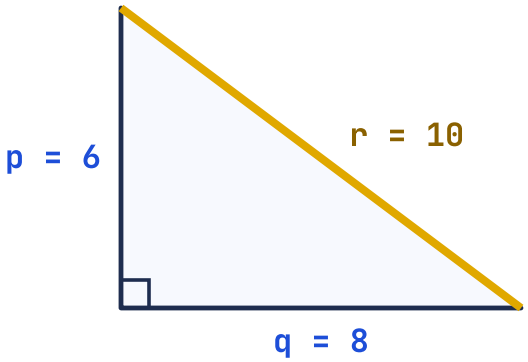
2. The hypotenuse is across from the right angle. Which side?

3. Check with the theorem: $8^2 + 15^2 = ?$

## Check your understanding

**1** Score 3 NC.8.G.6

Which side is the **hypotenuse**?




(A) r = 10

(B) p = 6

(C) q = 8

**2** Score 3 NC.8.G.6

A right triangle has legs a and b and hypotenuse c. Which equation is true?

(A) $a^2 \cdot b^2 = c^2$

(B) $a^2 + c^2 = b^2$

(C) $a^2 + b^2 = c^2$

(D) $a + b = c$

## 5.2 Find the hypotenuse NC.8.G.7

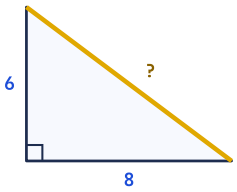
When the **hypotenuse is missing**: square each leg, **add**, then take the **square root**.

**Square root (√)**

undoes squaring: √100 = 10

**Round to the nearest tenth**

one digit after the decimal point: √13 ≈ 3.6




$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

$6^2 + 8^2 = c^2$

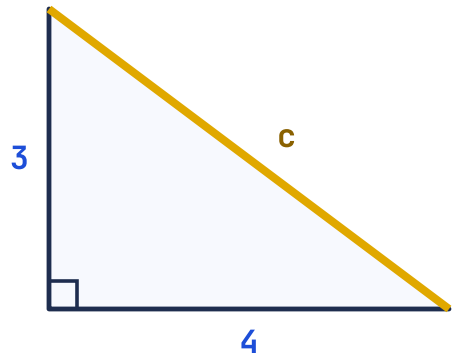
$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

**Watch out:** Do not add the legs (6 + 8 = 14). Square them first.

### Walk through: `Legs 3 and 4. Find the hypotenuse.`




1. Write the theorem with the legs in place.

_____

2. Square and add.

_____

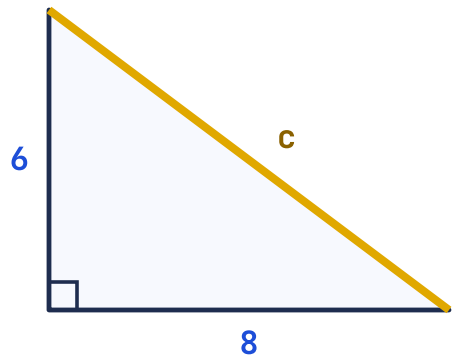
3. $c^2 = 25$. Find c.

_____

### Check your understanding

**1** Score 3 NC.8.G.7

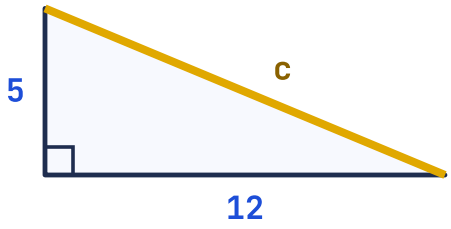
The legs of a right triangle are **6** and **8**. Find the hypotenuse.




Answer: _____

**2** Score 3 NC.8.G.7

The legs of a right triangle are **5** and **12**. Find the hypotenuse.




Answer: _____

## 5.3 Find a missing leg NC.8.G.7

When a **leg is missing**: square the hypotenuse and the leg you know, **subtract**, then take the **square root**. `leg² = c² - (other leg)²`

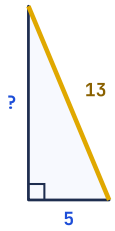
**Hypotenuse → add**

missing the long side: add the squares

**Leg → subtract**

missing a short side: subtract the squares

**Watch out:** If the missing side is a leg, the answer must be **shorter** than the hypotenuse. If it is not, you added by mistake.



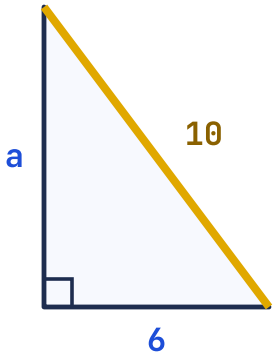

$a^2 + 5^2 = 13^2$

$a^2 + 25 = 169$

$a^2 = 169 - 25 = 144$

$a = \sqrt{144} = 12$

Walk through: **Hypotenuse 10, one leg 6. Find the other leg.**




1. Which side is missing?

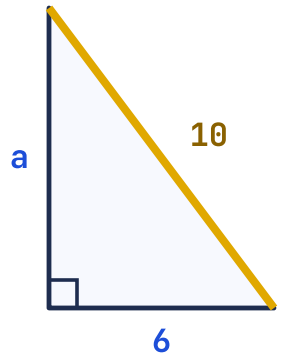
2. Square and subtract.

3. Find a.

### Check your understanding

**1** Score 3 NC.8.G.7

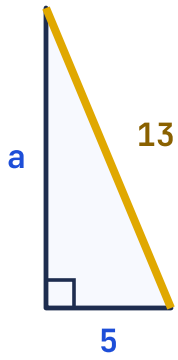
A right triangle has hypotenuse **10** and one leg **6**. Find the other leg.




Answer: _____

**2** Score 3 NC.8.G.7

A right triangle has hypotenuse **13** and one leg **5**. Find the other leg.




Answer: _____

## 5.4 Is it a right triangle? (the converse) NC.8.G.6

To test three side lengths: square the two **shorter** sides and add. If the sum **equals** the longest side squared, it **is** a right triangle. If not, it is not.

**Converse**
the theorem used backward, to test a triangle

**Longest side**
the only side that could be the hypotenuse

**Watch out:** Always use the **longest** side as c.

`Sides 6, 8, 10` `36 + 64 = 100` `100 = 100 ✓` right triangle

`Sides 4, 5, 6` `16 + 25 = 41` `41 ≠ 36 ✗` not a right triangle

### Walk through: `Do 9, 12, 15 make a right triangle?`

1. Which side would be the hypotenuse?

_____

2. Square and add the two shorter sides.

_____

3. $15^2 = 225$. What do we decide?

_____

### Check your understanding

**1** Score 3 NC.8.G.6

Can the sides **6, 8, 10** make a right triangle?

(A) Yes: 36 + 64 = 100, which is $10^2$

(B) No: the numbers are not 3, 4, 5

(C) No: 6 + 8 is not 10

**2** Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

(A) No: 16 + 25 = 41, but $6^2 = 36$

(B) Yes: 4 + 5 is more than 6

(C) Yes: they are in counting order

## 5.5 Distance on the coordinate plane NC.8.G.8

To find the distance between two points: draw a right triangle. The **across** change is one leg, the **up/down** change is the other leg, and the **distance** is the hypotenuse.

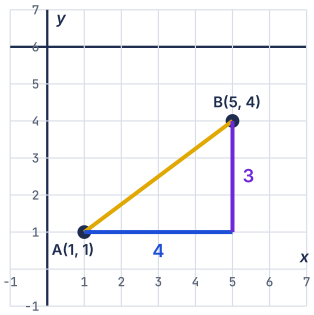
**Run**

the across change: subtract the x-values

**Rise**

the up/down change: subtract the y-values

**Watch out:** Count the length of each leg (always positive). Then use $a^2 + b^2 = c^2$.




```
run = 5 - 1 = 4
rise = 4 - 1 = 3
4² + 3² = 16 + 9 = 25
distance = √25 = 5
```

### Walk through: `Find the distance from (2, 1) to (8, 9).`

1 Find the run (across change).

2 Find the rise (up/down change).

3 Use $a^2 + b^2 = c^2$.

### Check your understanding

**1** Score 3 NC.8.G.8

Find the distance from **(1, 2)** to **(4, 6)**.

Answer: _____